

MANISH KUMAR

AI ENGINEER | COMPUTER VISION | SECURE ML | GenAI & LLM | HPC & RAG

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PROFESSIONAL SUMMARY

AI Engineer with 7+ years in technology and 4+ years specializing in AI/ML, with a proven track record of building production-grade computer vision and secure ML systems for DRDO (Ministry of Defence) and NHAI/MoRTH smart highway infrastructure. Deep expertise in real-time object detection (YOLO), edge deployment on NVIDIA Jetson/CUDA, and adversarial robustness testing (IBM ART). Recently completed NVIDIA training and currently configuring and managing NVIDIA H200 HPC GPU servers, including SLURM multi-node orchestration, group policy and access controls, and DDN 300TB high-performance storage for accelerated AI/ML and Generative AI model training. Strong infrastructure foundation from earlier IT administration roles. Actively expanding into Generative AI, RAG, and Agentic systems — including a production-ready PDF Chat Assistant. Seeking AI/ML or GenAI Engineer roles where production ML depth, HPC infrastructure experience, and growing LLM capabilities deliver measurable impact.

CORE TECHNICAL SKILLS

Computer Vision & Deep Learning: PyTorch, TensorFlow, CNNs, YOLOv4, OpenCV, Darknet, ResNet, MobileNet, VGG19, DenseNet

High-Performance Computing (HPC): NVIDIA H200 GPU Configuration & Management, CUDA, NVIDIA Jetson (Nano, Xavier NX, Orin NX), Multi-GPU / Multi-Node Training, Model Acceleration

HPC Infrastructure & Orchestration: SLURM Workload Manager, Node Configuration, Group Policy & Access Control, DDN 300TB High-Performance Storage

Secure & Explainable AI: IBM ART, Captum, Dioptra, MLflow, Adversarial Robustness (PGD, APGD, CW, FAB, AutoAttack), Model Evasion/Poisoning Testing

Deployment & Backend: Docker, FastAPI, Django, REST APIs, React.js, Linux, Git

Generative AI & LLMs: LLM Application Development, Prompt Engineering, Hugging Face, RAG, Conversational AI, Local LLM Deployment (Mistral, Gemma)

Programming & Data: Python, NumPy, Pandas, Shell Scripting

PROFESSIONAL EXPERIENCE

Software – AI Engineer

Oct 2024 – Present

GVR Technolabs Pvt. Ltd. (DRDO – Ministry of Defence Projects) | New Delhi

- Conducted comprehensive AI/ML security assessments using IBM ART, Dioptra, and Captum, focusing on adversarial robustness, explainability, and model trustworthiness for defence-critical applications.
- Performed evasion and poisoning attacks (PGD, APGD, CW, FAB, AutoAttack) against ResNet, MobileNet, VGG19, and DenseNet architectures on ImageNet to evaluate and harden model resilience.
- Evaluated security and safety of Generative AI systems, including Hugging Face models and locally deployed LLMs (Gemma-4 12B); designed automated testing and validation workflows.
- Built robust experimentation and tracking pipelines using MLflow to monitor model accuracy and performance under adversarial conditions.
- Configured and managed NVIDIA H200 HPC GPU servers for high-throughput AI/ML and Generative AI model training, delivering significantly accelerated training cycles and improved resource utilization.
- Implemented group policies, user access controls, and permission frameworks to ensure secure and controlled usage of shared HPC resources across teams.
- Deployed and administered SLURM workload manager across multi-node clusters for efficient job scheduling, resource allocation, and large-scale distributed training pipelines.
- Managed DDN 300TB high-performance storage infrastructure, ensuring high availability, fast data access, and reliable storage for large AI datasets and model checkpoints.

R&D Engineer – AI/ML

Aug 2022 – Oct 2024

VaaaN Infra Pvt. Ltd. (NHAI/MoRTH Smart Highway Projects) | Faridabad

- Developed and optimized YOLO-based vehicle detection and classification models for smart highway surveillance systems under NHAI/MoRTH mandate.
- Built custom training datasets from live highway camera feeds; trained and fine-tuned models using YOLO, OpenCV, and Darknet.
- Deployed real-time AI inference pipelines on NVIDIA Jetson (Nano, Xavier NX, Orin NX) and GPU-based server environments for continuous incident detection.
- Designed and delivered live monitoring and analytics dashboards using React.js, Django REST Framework, and REST APIs, actively used by highway operations teams.

- Managed Linux/Windows server infrastructure, networking, and GPU workstations — foundational systems experience later applied to AI model deployment; administered network security and CCTV/edge surveillance devices.

KEY PROJECTS

PDF Chat Assistant

- Built an LLM-powered PDF Question-Answering application using Python, Flask, Mistral-7B, and llama-cpp-python, enabling document-based conversational AI with RAG capabilities.

VIDS — Video Incident Detection System

- Real-time vehicle detection and incident-monitoring system using YOLO, OpenCV, and Darknet; deployed on NVIDIA Jetson edge devices for continuous highway surveillance.

ATCC — Automatic Traffic Counter & Classifier

- AI-powered traffic counting and vehicle classification system for NHAI/MoRTH highways; trained with YOLO/Darknet and deployed on CUDA-enabled GPU edge devices.

EDUCATION

Master of Computer Applications (MCA)

Swami Vivekanand Subharti University

2022

B.Sc. Information Technology

Maharishi University

2018

CERTIFICATIONS

Neural Networks & Deep Learning (DeepLearning.AI) | Python for Data Science & AI (IBM) | AWS DevOps Engineer – Professional | Linux Administration | NVIDIA GPU Computing / HPC Training